

VIT-AP
UNIVERSITY

DIGITAL LOGIC DESIGN
(Course Code: ECE 1003)

Module-1:Lecture-7
Boolean Algebra

CONTENTS

Module-1 (Part-2)

- ❖ **Logic gates and universal gates**
- ❖ Tri-state logic
- ❖ Boolean algebra
- ❖ SOP and POS minimization
- ❖ K-maps upto 4-variables,
- ❖ Don't care logic

CONTENTS

Lecture-7

- ❖ Binary Logic
- ❖ Logic Gates

BOOKS

Textbooks

1. M.Morris Mano, Michael D Ciletti, **Digital Design**, 5th edition, Pearson Publishers, 2013.
2. R.P. Jain, “**Modern Digital Electronics**”, 4th edition, TMH.

References

1. M.Morris Mano, Charles R. Kime, Tom Martin, **Logic and Computer Design Fundamentals**, 4th edition, Pearson Publishers.
2. C. H. Roth and L. L. Kinney, **Fundamentals of Logic Design**, 5th edition, Cengage Publishers.

Binary Logic & Logic Gates

BINARY LOGIC

- ✓ **Binary logic** is used in order to have an **abstract representation of logic gates**.
- ✓ It has **two discrete values** such as YES or NO, TRUE or FALSE, LOW or HIGH, 0 or 1, etc.
- ✓ A digital logic circuit has several input lines A, B, C, ... and several output lines. We will consider one output line called Z at the moment.
- ✓ We write $Z = f(A, B, C, \dots)$ to mean that the value of Z is determined by the values of the inputs A, B, C.... and Z is said to be a function of its inputs.
- ✓ In the case of real digital circuits, currents are passed through the lines, but voltages between about 5 and 15 are called **ON or TRUE** and near zero called **OFF or FALSE**.
- ✓ We identify TRUE with the digit 1 and FALSE with the digit 0.
- ✓ There are 3 basic logical operations AND, OR and NOT.

BINARY LOGIC

AND OPERATOR

It performs logical multiplication and denoted by (.) dot.

X	Y	X.Y
0	0	0
0	1	0
1	0	0
1	1	1

OR OPERATOR

It performs logical addition and denoted by (+) plus.

X	Y	X+Y
0	0	0
0	1	1
1	0	1
1	1	1

NOT OPERATOR

It performs logical negation and denoted by bar. It operates on single variable.

X	$\overline{X}$
0	1
1	0

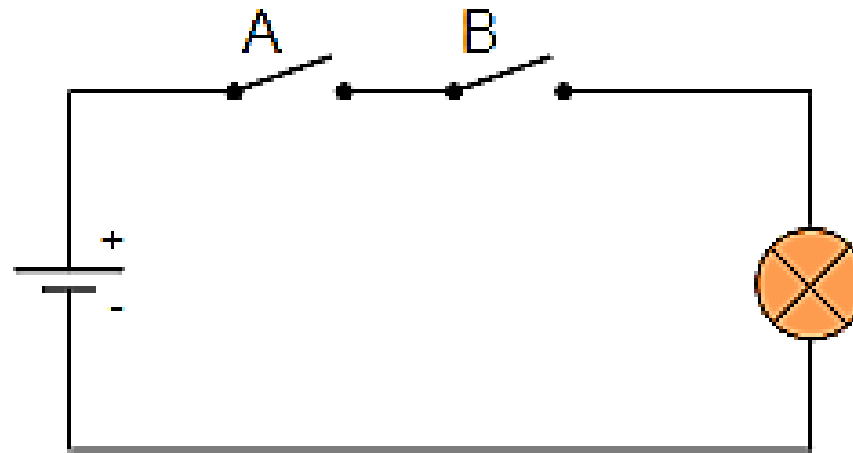
(means complement of x)

BINARY LOGIC

AND OPERATOR

It performs logical multiplication and denoted by (.) dot.

X	Y	X.Y
0	0	0
0	1	0
1	0	0
1	1	1



Lamp - ON = "1"
Lamp - OFF = "0"

Switch A - Open = "0", Closed = "1"

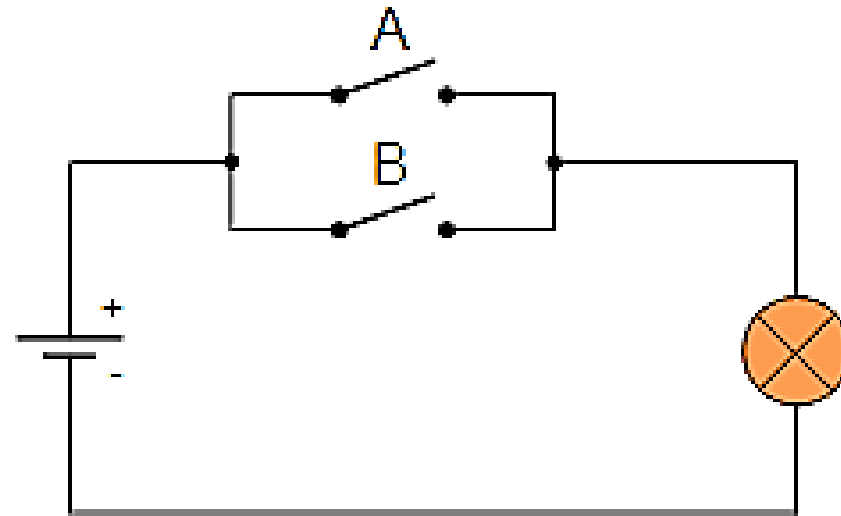
Switch B - Open = "0", Closed = "1"

BINARY LOGIC

OR OPERATOR

It performs logical addition and denoted by (+) plus.

X	Y	X+Y
0	0	0
0	1	1
1	0	1
1	1	1



Switch A - Open = "0", Closed = "1"

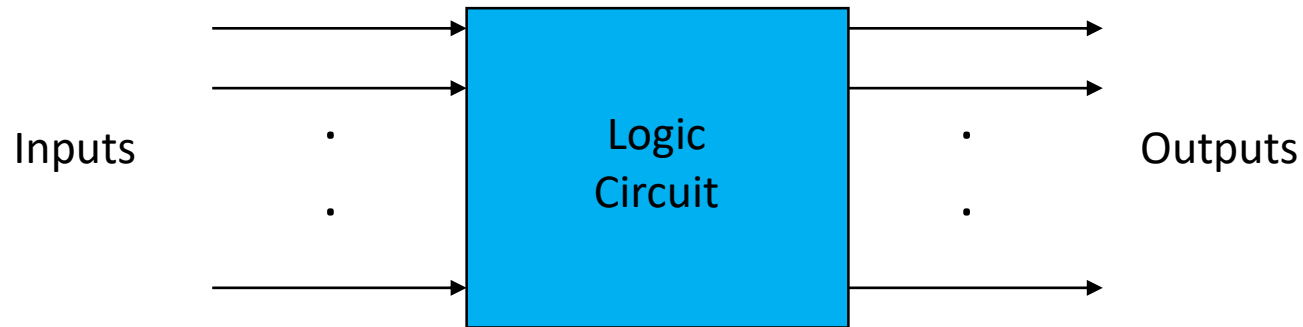
Switch B - Open = "0", Closed = "1"

Lamp - ON = "1"

Lamp - OFF = "0"

DIGITAL SYSTEMS

✓ Analysis problem:

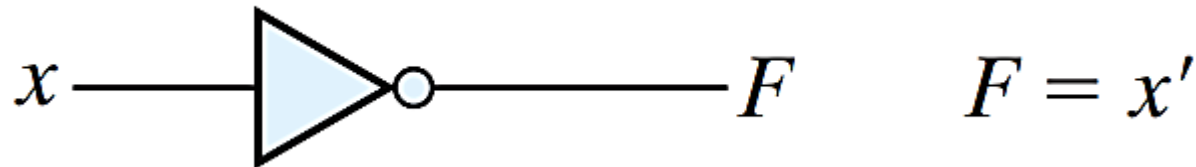


- Determine binary outputs for each combination of inputs
- ✓ Design problem: given a task, develop a circuit that accomplishes the task
 - Many possible implementation
 - Try to develop “best” circuit based on some criterion (size, power, performance, etc.)

LOGIC GATES

Describing Circuit Functionality: Inverter or NOT gate

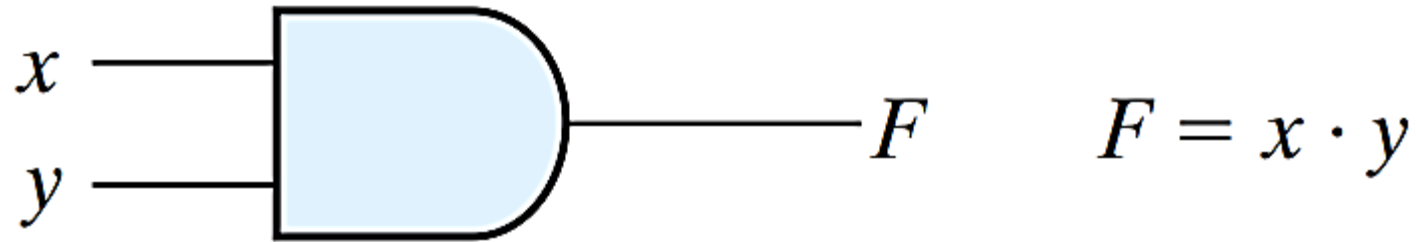
- ✓ Basic logic functions have symbols.
- ✓ The same functionality can be represented with truth tables.
 - Truth table completely specifies outputs for all input combinations.
- ✓ The below circuit is an inverter or NOT gate.
 - An input of 0 is inverted to a 1.
 - An input of 1 is inverted to a 0.



x	F
0	1
1	0

LOGIC GATES

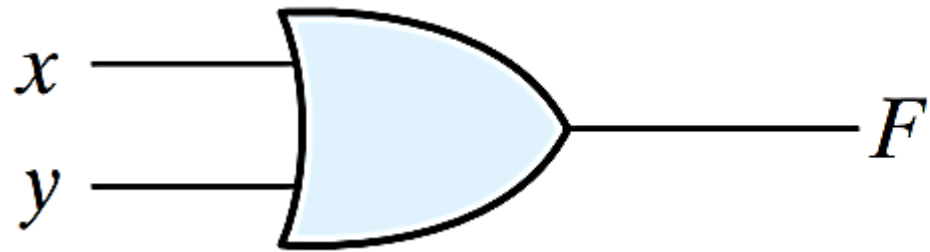
Describing Circuit Functionality: **AND** Gate



x	y	F
0	0	0
0	1	0
1	0	0
1	1	1

LOGIC GATES

Describing Circuit Functionality: **OR** Gate

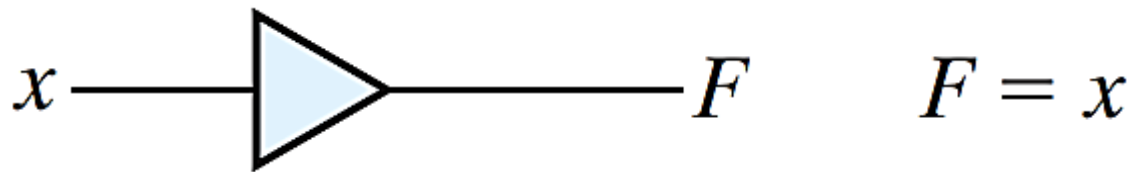


$$F = x + y$$

x	y	F
0	0	0
0	1	1
1	0	1
1	1	1

LOGIC GATES

Describing Circuit Functionality: **Buffer**

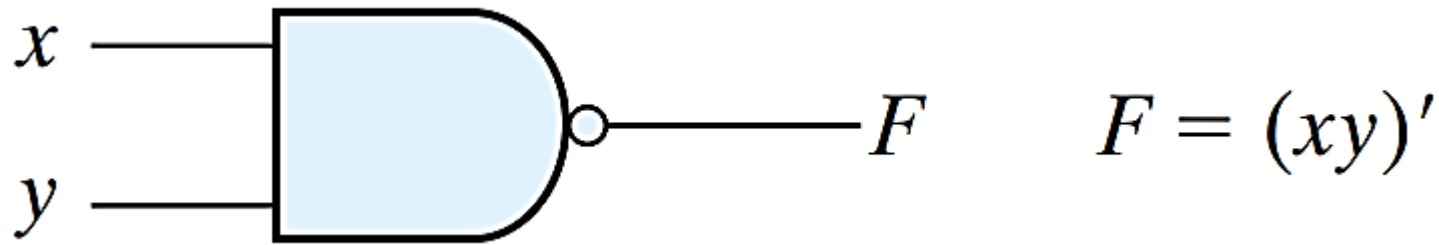


x	F
0	0
1	1

LOGIC GATES

Describing Circuit Functionality: **NAND** Gate

✓ This is called as Universal Gate.

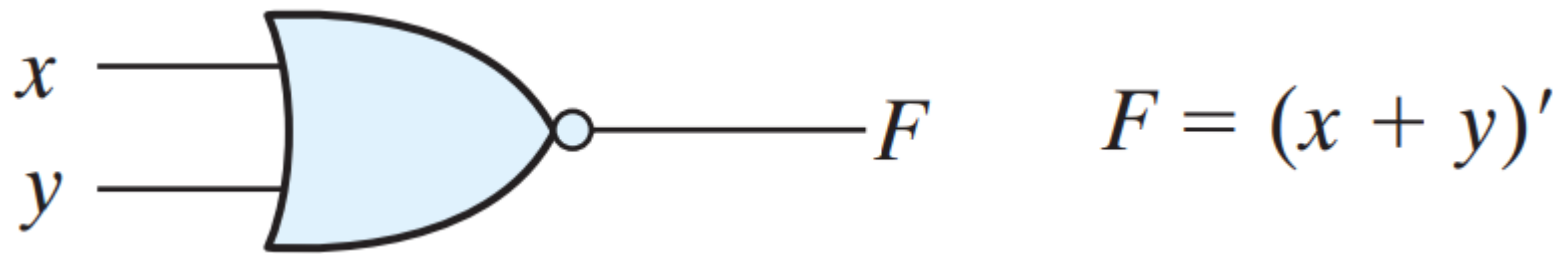


x	y	F
0	0	1
0	1	1
1	0	1
1	1	0

LOGIC GATES

Describing Circuit Functionality: **NOR** Gate

✓ This is also called as Universal Gate.

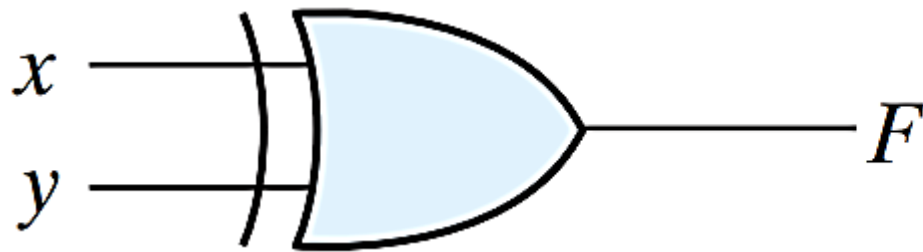


x	y	F
0	0	1
0	1	0
1	0	0
1	1	0

Note: Any logical function can be implemented using these Universal gates, i.e., all other gates can be implemented using Universal Gates.

LOGIC GATES

Describing Circuit Functionality: **EX-OR** or **XOR** Gate

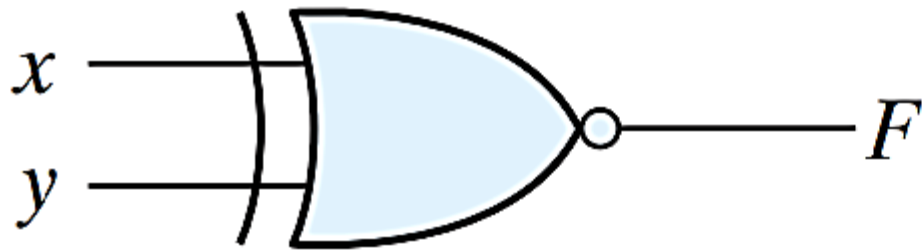


$$F = xy' + x'y$$
$$= x \oplus y$$

x	y	F
0	0	0
0	1	1
1	0	1
1	1	0

LOGIC GATES

Describing Circuit Functionality: **EX-NOR** Gate



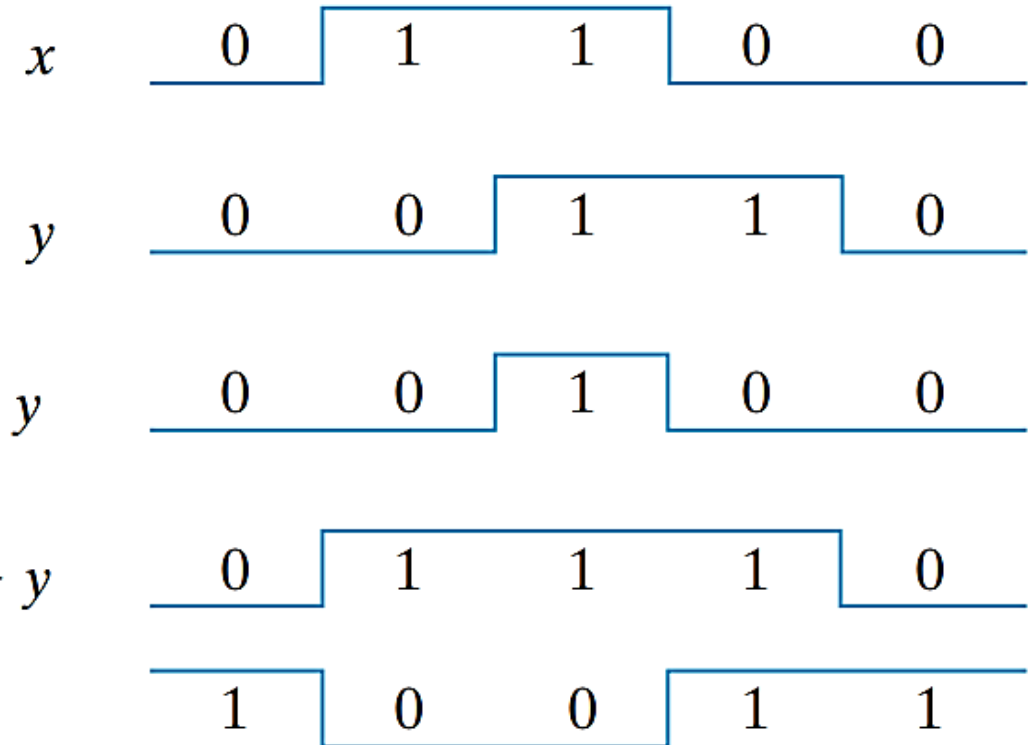
$$F = xy + x'y' \\ = (x \oplus y)'$$

x	y	F
0	0	1
0	1	0
1	0	0
1	1	1

LOGIC GATES

Describing Circuit Functionality: Wave form

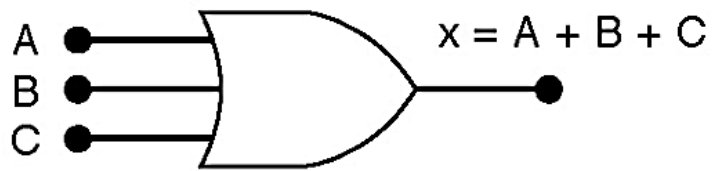
- ✓ Waveforms provide another approach for representing functionality.
- ✓ Values are either high (logic 1) or low (logic 0).
- ✓ Can you create a truth table from the waveforms?



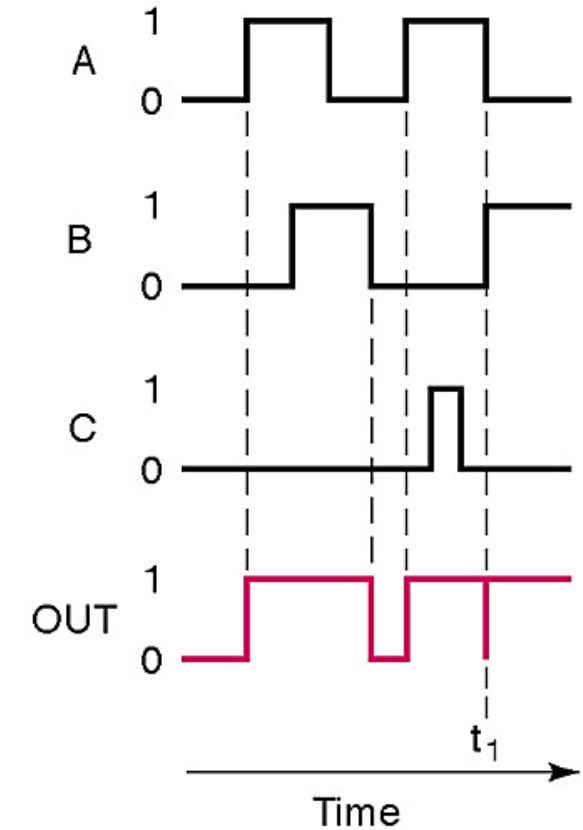
LOGIC GATES

Multi Input Gates

3 Input OR Gate



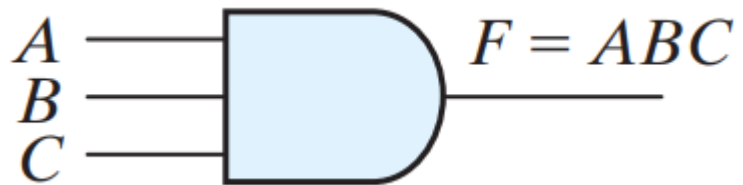
A	B	C	$x = A + B + C$
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1



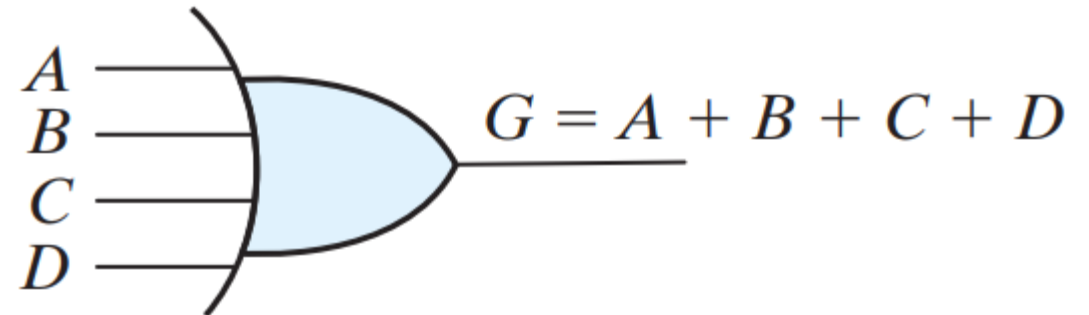
LOGIC GATES

Multi Input Gates

3 Input AND Gate



4 Input OR Gate



Thank You